LAB REPORT

COURSE INFORMATION

- Course Code:
- Course Title:
- Experiment No:
- Experiment Name:

SUBMITTED TO

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- Department of
- East West University

SUBMITTED BY

- Student Name :
- ID :
- Section :
- Department :

East West University

Date of submission:

DATA SHEET: Measurement of DC and AC voltages and determination of line frequency using an oscilloscope and a function generator.

Table -1: Determining **DC voltage**:

No. of Obs.	Volts per Division A *Volts*	Number of Divisions Deflected B	EMF of the Battery $A \times B$ *Volts*	Mean EMF *Volts*
1				
2				
3				

Table -2: Determining **AC voltage**:

No. of Obs.	Volts per Division A *Volts*	Number of Divisions from top to bottom B	Peak to Peak Voltage of the signal $A \times B$ *Volts*	Mean Peak to Peak Voltage, V_{p-p} *Volts*
1				
2				
3				

Table -3: Determining **line Frequency**:

No. of Obs.	Sweep time per Division A *ms*	Number of Divisions for one wave length B	Time period T $A \times B$ *ms*	Frequency of the signal $f = \frac{1}{T}$ *Hz*	Mean Frequency *Hz*
1					
2					
3					

Table -4: Determining **line Frequency (f_x) by Lissajous figure**:

No. of Obs.	Known frequency from function generator f *Hz*	Number of vertical loops L_V	Number of horizontal loops L_H	Line frequency $f_X = \frac{L_H}{L_V} \times f$ *Hz*	Mean Frequency *Hz*
1					
2					
3					
4					

Calculation:

$$\text{Percentage Difference} = \frac{\text{Standard value} \sim \text{Experimental value}}{\text{Standard value}} \times 100\%$$

Experimental value of **DC voltage** (measured by oscilloscope experiment) =…………

Standard value of DC voltage (measured by multimeter) = ……………

Percentage difference = …………….

Experimental value of **AC voltage** (measured by oscilloscope experiment), $\mathbf{V_{p\text{-}p}}$ =……………..

AC voltage measured by multimeter, $\mathbf{V_{rms}}$ **=**

Standard value of AC voltage (measured by multimeter), $V_{p-p} = 2.\sqrt{2}.V_{rms} =$ ……… …..

Percentage difference = …………….

Experimental value of **AC frequency** (measured by Lissajous figure) =……………..

Standard value of AC frequency = ……**50**…**Hz**……. (Known)

Percentage difference = …………….

Results:

(i) DC voltage =

(ii) AC voltage =

(iii) Line frequency (measured by Time/Div knob) =

(iv) Line frequency (measured by Lissajous figure) =

Discussions:

(i) As mentioned in the procedure, it is better to have the Lissajous pattern slowly rotating.

(ii) It is difficult to record the number of both the horizontal and vertical loops at the same time. It is better to fix the number of either vertical or horizontal loops constant and then vary the number or the other and note the corresponding frequency of the signal from the signal generator.